Pico 2 mini OS Markdown Viewer

Pico 2 mini OS · October 30, 2026

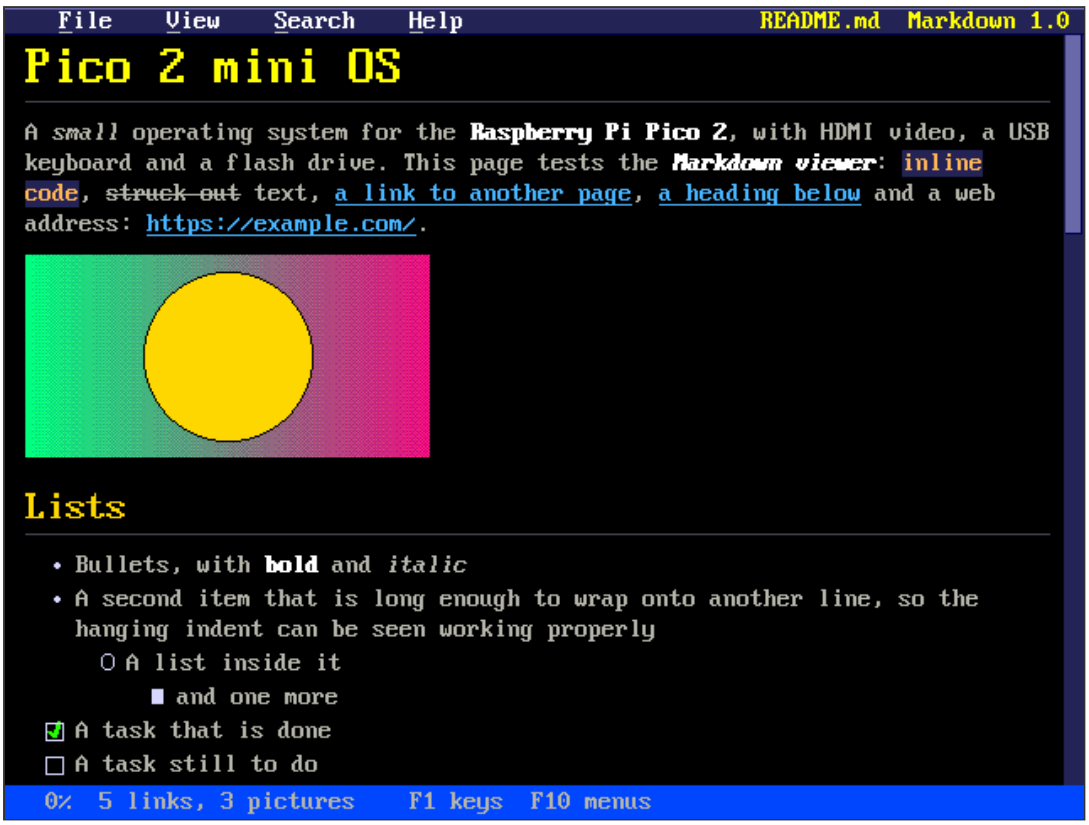


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`md` shows Markdown files from the drive in graphics: big headings, bold and italic text, code on a dark band, quotes, lists, task lists, tables, rules, links you can follow and pictures (PNG, JPEG and BMP) from the drive in their place. READMEs, notes and the manuals in `docs/` read well on it.

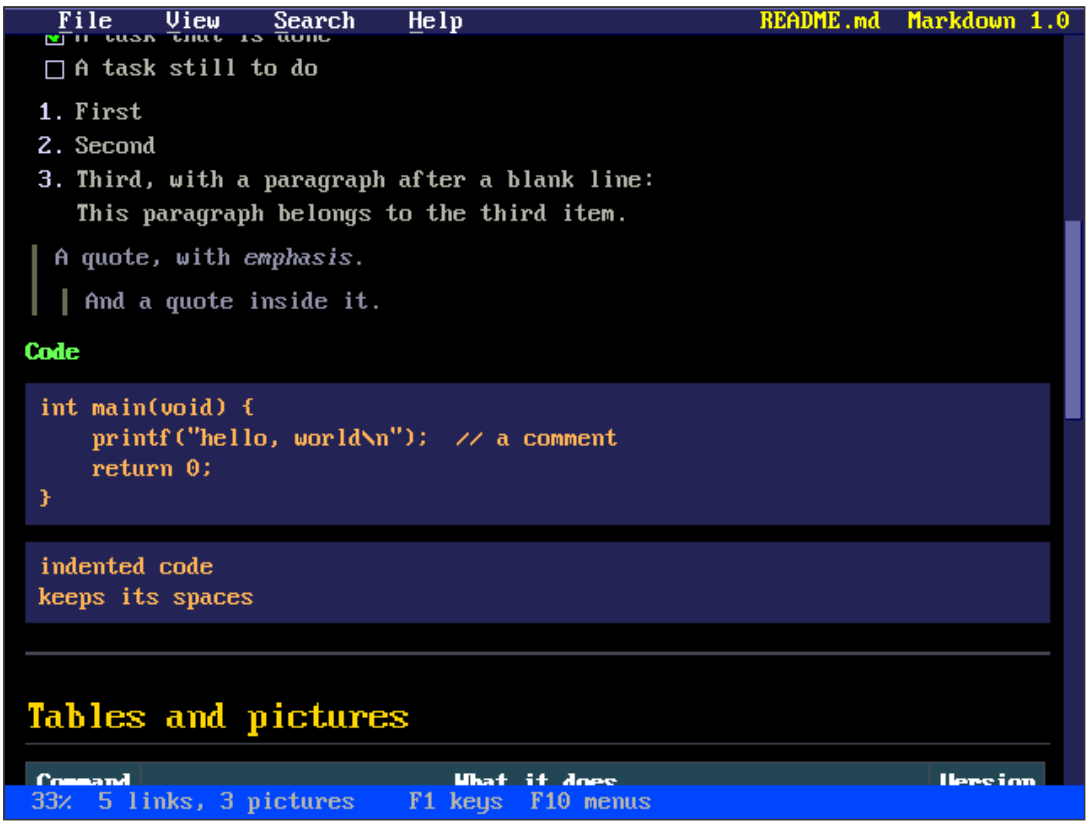
The viewer



`md README.md` (or `markdown README.md`) opens a file; `md` alone opens the file box in the shell's folder, and `md /DOCS` opens it in that folder. In `mc`, Enter on a `.md` file opens it here. It works with the keyboard alone; a USB mouse helps.

- **The menu bar** at the top: File, View, Search and Help, as in the tracker and the hex editor. Click a title, or press F10 (or F9), or Alt and the title's first letter. In a menu, Up and Down choose, Left and Right go to the next menu, Enter or the underlined letter runs an item, Esc closes it. The file's name and the viewer's version are on the right.
- **The page**, 75 characters wide. Up and Down scroll, PgUp, PgDn, Space and `b` go a page, Home and End to the top and the end; the wheel scrolls too.
- **The scroll bar** on the right: drag the thumb, or click above or below it for a page.
- **The status line** at the bottom: how far down the page is, how many links and pictures it has, the address of the link under the pointer (or the one chosen), or a message.

What it shows



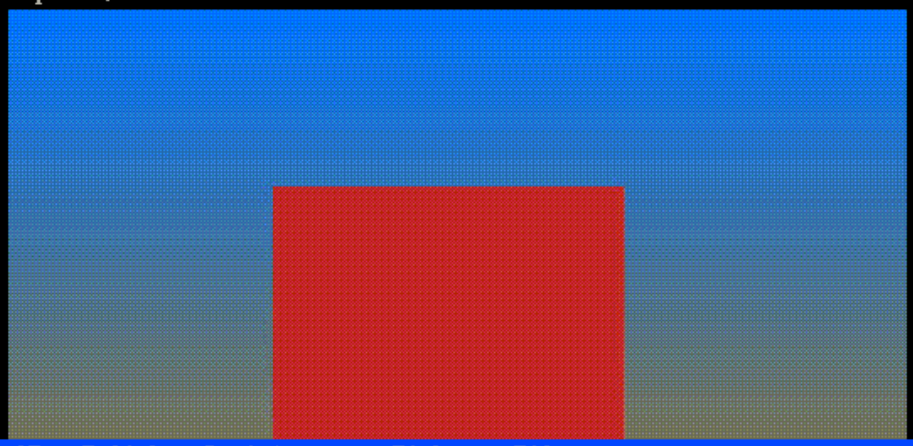
- **Headings:** `#` (and a line of `===` under the text) in 16x32 letters with a rule under them, `##` (and `---`) in 12x24, `###` to `######` in bold.
- **Text:** `*italic*` and `_italic_` slanted, `**bold**` in thicker, white strokes, `~~struck out~~`, `` `code` `` on a dark background. Lines join into paragraphs; a line ending in two spaces or `\` breaks there. A few HTML tags work too: `<b>`, `<i>`, `<code>`, `<br>`, `<img>` and `<a href>`; others are left out. Entities (`&`, `©`, `é` ...) and UTF-8 are shown in the 8x16 font's code page 437, accents and all where it has them.
- **Lists:** bullets (`-`, `*`, `+`), numbers (`1.` or `1)`), one inside another, and task lists (`- [ ]` and `- [x]`, a box and a tick).
- **Quotes** (`>`) with a bar on the left, one inside another.
- **Code** between fences (```` ``` ```` or `~~~`) or indented four spaces: as it is, on a dark band, cut where a line is too long.
- **Tables** (GitHub's: a `|---|:---:|---:|` row under the header) with their columns' alignment, the header row shaded and every other row too. Columns too wide for the page are narrowed and their text wrapped.
- **Rules** (`---`, `***`, `___`).
- **Pictures:** `![alt text](picture.png)` and `<img src=... width=...>` are drawn in their place, found from the Markdown file's folder (or the drive's root, for a name starting with `/`). PNG, baseline JPEG and BMP files work; one wider than the page is shrunk to fit, and none is taller than 400 pixels. Each is drawn the first time it comes on the screen ("Drawing ..." on the status line; a big PNG takes a few seconds). Pictures from the web, and ones that can't be shown, are their alt text in pink, `[like this]`. `*` (View > Pictures) shows boxes instead.

File View Search Help README.md Markdown 1.0

Tables and pictures

Command	What it does	Version
md	Shows Markdown	1.0
hex	Edits bytes in hex	1.0
www	Browses the web, in text, with pictures in pages when asked for	1.0

A photo, scaled to fit:



65% 5 links, 3 pictures F1 keys F10 menus

TrueType text



If there are fonts in `/FONTS` on the drive, `md` draws its text with them: proportional and anti-aliased, with real bold and italic letters. Code, code blocks, the source view and list bullets stay in the 8x16 font, so code still lines up. F7 or `f` (or View > TrueType text) switches between the fonts and the 8x16 font.

The fonts drawn are:

Style	File (the default)
regular	`/FONTS/SS3.TTF`
bold	`/FONTS/SS3-B.TTF`
italic	`/FONTS/SS3-I.TTF`
bold italic	`/FONTS/SS3-BI.TTF`

Only the regular font is needed. Without a bold font, bold is drawn as the regular font twice, one pixel apart. Without an italic font, italic text is upright. To use other fonts, write `/FONTS/MD.CFG` with lines like these (a name without a `/` is in `/FONTS`):

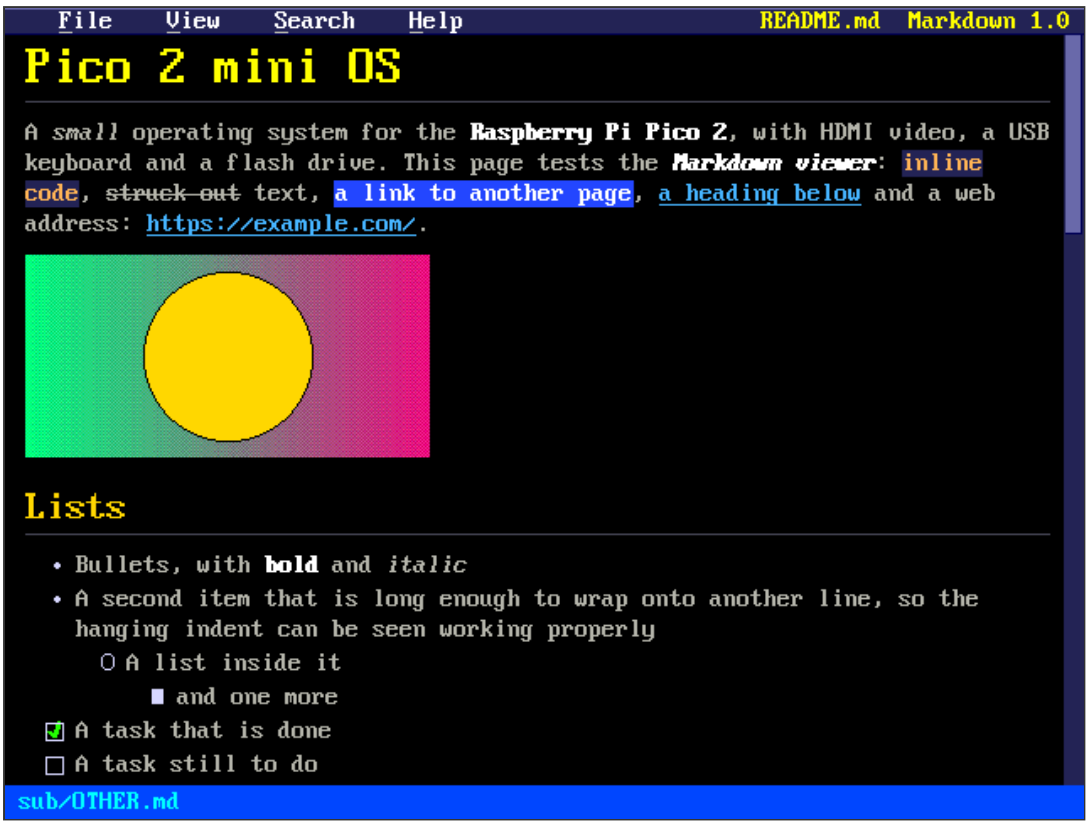
```
regular=DejaVuSans.ttf
bold=DejaVuSans-Bold.ttf
italic=/MYFONTS/DejaVuSans-Oblique.ttf
```

Any font the `font` command reads will do: `.ttf`, `.otf` or `.pfn` (see the *Font Viewer* manual). Body text is drawn at 16 pixels per em, `##` headings at 24 and `#` headings at 32. Lines keep

the 8x16 font's heights. Each glyph is drawn once and kept in a cache, so scrolling redraws from the cache. Kerning is not used.

Loading takes longer with big font files. Source Sans 3 cut down to the characters the viewer can show is about 19 KB per style, against 430 KB for the whole font. To make such a file on a PC with fontTools, run `pyftsubset SourceSans3-Regular.ttf --unicodes=U+20-7E,U+A0-FF --no-hinting --output-file=SS3.TTF`. The fonts are kept in PSRAM, with a 16 KB working area in SRAM, only while `md` is open. If there are no fonts, or not enough memory, the 8x16 font is used.

Links



Links are blue and underlined. Tab chooses the next one (Shift-Tab the one before), starting from the top of the screen, and the page scrolls to follow; Enter (or Right) follows the one chosen. A click follows a link at once. What following does depends on where it goes:

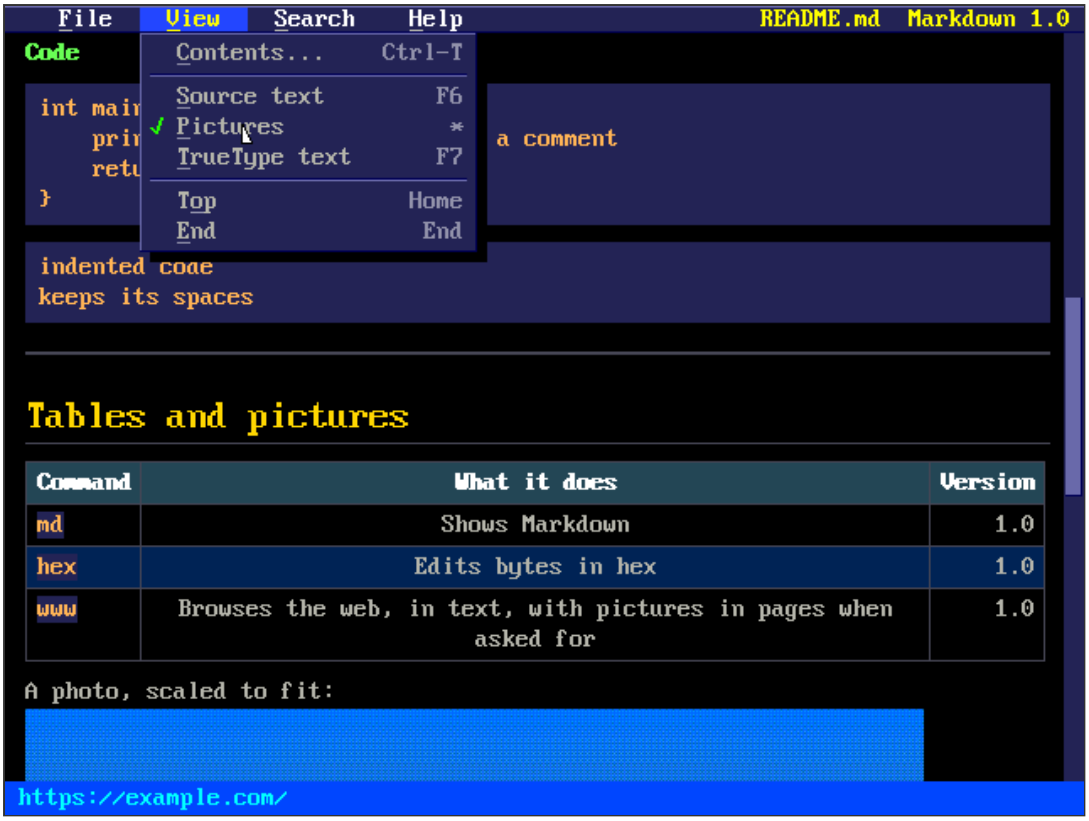
- **Another Markdown file** (`[next](CHAPTER2.md)`, `[up](../README.md)`) opens here. Backspace (or Left, or `[`) goes back to the file before, at the place it was left; `]` goes forward again. A `#name` after the file goes to that heading in it.
- **A heading** in this file: `[see below](#tables-and-pictures)`. Its name is made as GitHub makes them: the heading's text in lower case, spaces as `-`, other signs left out.
- **A picture** (`.png`, `.jpg`, `.bmp`) fills the screen; any key or click comes back. A click on a picture in the page does the same.
- **A folder** opens the file box there.
- **A web address** (`http://` or `https://`) asks first, then opens in the web browser, `www`; quitting it comes back here.

Reference links work too: `[text][name]`, `[name][]` and `[name]`, with `[name]: address` anywhere in the file.

Contents, Find and the source



- **Ctrl-T** (or `t`, View > Contents) lists the headings, indented by level, with the one the page is at chosen. Up, Down, PgUp and PgDn choose; Enter (or a click) goes there.
- **Ctrl-F** (or `/`) finds text (case doesn't matter) from the top of the screen down; it is shown on yellow. F3 (or `n`) finds the next.
- **F6** (or `s`, View > Source text) shows the file as it is, each line wrapped; F6 again lays it out once more.
- **F5** (or `r`) reads the file again, keeping the place: handy while editing it in `edit` or on a PC.
- **Ctrl-O** (or `o`) opens another file: the file box lists the folder's folders and files (Markdown files in blue); Enter or a double click opens one, Backspace goes up a folder.



The right mouse button gives a menu of its own: follow the link under the pointer, Back, Forward, Contents, Find, Source text and Reload.

Keys

Key	What it does
Up, Down	scroll
PgUp, PgDn, Space, b	a page up or down
Home, End	the top, the end
Tab, Shift-Tab	choose the next or the previous link
Enter, Right	follow the link chosen
Backspace, Left, `[`	back
`]`	forward
Ctrl-T, `t`	contents
Ctrl-F, `/`	find
F3, `n`	find the next
F6, `s`	the source text, or the page
`*`, `p`	pictures, or boxes
F7, `f`	TrueType text, or the 8x16 font
F5, `r`	read the file again
Ctrl-O, `o`	open a file
F1, `?`	the keys
F10, F9	the menus
Esc, `q`	quit

Limits

- Each frame is drawn 16 rows at a time in a 10 KB strip of SRAM (only while `md` is open) and copied onto the screen piece by piece: about 30 ms a frame with the 8x16 font and 47 ms with TrueType text, against 120 and 142 ms through a picture in PSRAM. Ctrl-B switches between the two, and About gives the times.
- Pictures are decoded when they come into view. A PNG without transparency is decoded a row at a time, straight into the picture kept for the page, never whole: a 640x480 screenshot takes about a third of a second, against 1.9 s before. While one decodes, `md` lends it its strip's and the fonts' SRAM. About gives the pictures' times.
- Files up to 8 MB. The text, the page laid out and the pictures drawn (up to 3 MB of them; those furthest from the screen are let go first and drawn again when they come back) are kept in PSRAM, only while `md` is open.
- Pictures from the web are not fetched; GIF, SVG and progressive JPEGs are shown as their alt text.
- HTML blocks are shown as their text, with the tags left out; footnotes and math are shown as they are written.
- Text is kept as code page 437, in both fonts: characters it lacks (most of those outside Western European languages, and emoji) are left out.